

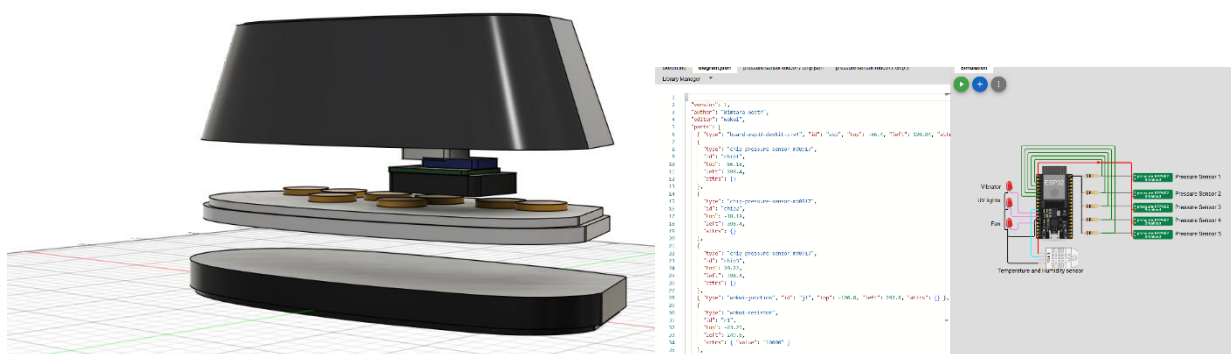
INTENSE CARING SHOE FOR DIABETIC PATIENTS

TECHNICAL PROGRESS REPORT

Platform-ESP 32 Platform , Project Phase – Semi Final progress & System Validation

Project Implementation Road Map

1. Phase 1 – Architecture and simulation Setup (Completed)
 - 1.1. Wokwi Simulation Setup(ESP32 , 5X FSR Sensors , DHT22 , Actuator)
 - 1.2. Pin Mapping , ADC Voltage Dividers & Digital Signals
 - 1.3. Local Web Access Point(192.168.4.1)& Data Endpoint Design
2. Phase 2-System Firmware & Multi- Sensor Integration (Current Phase)
 - 2.1. 5-point Plantar Pressure Array Signal Processing & Calibration
 - 2.2. Temperature , Humidity and pH Microclimate Alert Threshold
 - 2.3. Haptic Offline Alerts & Actuator Command Engine
3. Phase 3- Physical Prototyping & Mechanical Assembly (Next Step)
 - 3.1. Flexible Insole Fabrication & Pressure Sensor Matrix Mounting
 - 3.2. Mechanical Protection Enclosure for Control Unit & Wiring Routing
 - 3.3. Power Electronics & Transistor Switching Drivers
4. Phase 4- Field Validation & Final Competition Defense
 - 4.1. AI integration
 - 4.2. Connecting to a cloud dashboard



Screenshot 1- 3D model of the shoe made with AutoDesk Fusion

Screenshot 2- Online simulation of the circuit made with Wokwi

The following table shows the hardware pin mapping and circuit specifications

Component	Part ID /Pins	ESP32 GPIO	Signal Type
Pressure Sensor(Hallux)	Chip 1	GPI036(VP)	Analog Input via Omega Voltage divider
Pressure Sensor 2(1st Metatarsal)	Chip 2	GPI039(VN)	Analog Input via Omega Voltage Divider
Pressure Sensor 3(3rd Matatarsal)	Chip 3	GPI034	Analog Input via Omega Voltage Divider
Pressure Sensor 4(5th Metatarsal)	Chip 4	GPI035	Analog input via Omega voltaje divider
Pressure Sensor 5(Heel)	Chip 5	GPI032	Analog input Omega Voltage Divider
Temp&Humidity Sensors	dht 1	GPI027	One-wire Digital interface(DHT22)
Haptic vibration Actuator	led2	GPI014	Digital Output(High pressure Warning)
UV-C Sterilization LED	led3	GPI012	Digital output (Microbial Control)
Cooling Fan Actuator	led4	GPI013	Digital Output(Thermal control)